A spherical version of the axisymmetric shallow-water model

Plan for `spencerahill/axisym-hc-swm`

Prepared 2026-11-07, and revised the same day to Spencer's answers on the setup and his choices of the two open terms. Re-derived on 2026-11-08 against the model's `main`, after the merge of moist V2. This copy covers the model work, phases 1 to 4; the science runs that follow are planned in the private repository `spencerahill/fixed-ro-eq-area`. Symbolic and discrete checks, and quoted numbers, from `scripts/spherical_swm_plan_checks.py` in that repository.

1 Executive Summary

The axisymmetric shallow-water model (SWM) of the Hadley cell's upper branch in `spencerahill/axisym-hc-swm`, after Sobel and Schneider (2009, hereafter SS09), runs on an equatorial beta plane. The SWM's Coriolis parameter is βy, and its equations contain no metric terms and no planetary radius, so they are exactly the small-angle limit of the spherical equations. The SWM therefore cannot test a theory of the Hadley cell beyond the small-angle limit. This plan adds spherical geometry to the SWM as an option, with every beta-plane run left bit-for-bit unchanged.

Where the spherical equations come from. SS09 adapted their model from Xian and Miller (2008), who wrote the same two-layer axisymmetric equations on the sphere, pole to pole, in the coordinate $\sin\varphi$ for latitude φ; Adam and Paldor (2009) used the same coordinate. The spherical SWM takes its dynamics from Xian and Miller and writes SS09's additional terms on the sphere. The spherical SWM's coordinate is $y = a\sin\varphi$, for planetary radius a, with $c = \cos\varphi = (1 - y^2/a^2)^{1/2}$. This y is a coordinate, not meridional distance, which is $a\,d\varphi = dy/c$. In it the Coriolis parameter is $f = 2\Omega\sin\varphi = (2\Omega/a)\,y$ exactly, for rotation rate Ω. The constant $2\Omega/a$ is the equatorial value of the physical $\beta = 2\Omega\cos\varphi/a$, which varies with latitude. The upper-layer equations are

$$\frac{\partial u}{\partial t} - v\left(f - \frac{\partial (cu)}{\partial y}\right) = -\mathcal{G}\,u\,\frac{\partial (cv)}{\partial y} - \epsilon_u u - \mathcal{S}, \tag{1}$$

$$2\frac{\partial v}{\partial t} + \left(f + \frac{u\tan\varphi}{a}\right)u = -\frac{gH}{T_0}\,c\,\frac{\partial T}{\partial y} + \mathcal{K}_v, \tag{2}$$

$$\frac{\partial \theta}{\partial t} + \frac{\delta\Delta_z}{H}\,\frac{\partial (cv)}{\partial y} = \frac{\theta_E - \theta}{\tau}, \tag{3}$$

with $\tan\varphi/a = y/(a^2 c)$. Here $\mathcal{G}$ is the existing gate on the vertical advection of zonal momentum, ϵ_u the Rayleigh drag coefficient, $\mathcal{S}$ the parameterized eddy momentum flux divergence, and $\mathcal{K}_v$ the smoothing of v. Setting $c = 1$, dropping the metric term and writing $f = \beta y$ recovers the current equations symbol for symbol. So the sphere differs from the beta plane through the factors of c in three derivatives, the metric term, and the domain, which ends at the poles, $y = \pm a$. Section 2 goes through every term.

Two properties make the spherical SWM a test of full-angle theory. The angular-momentum-conserving (AMC) wind $\Omega a\sin^2\varphi/\cos\varphi$ is an exact steady state of the inviscid form of Eq. (1). And the steady form of Eq. (2) is the full-angle gradient wind balance. These properties, the coordinate identities, the beta-plane limits and the properties claimed in Section 2 were checked symbolically, with 20 of 20 checks passing. The discrete properties claimed in Section 3 were checked numerically against prototype operators built from the model's own functions, with 9 of 9 checks

passing, both at the commit the plan was first written against and at the merged `main`, commit `0dc0627`.

Settled, from Spencer's answers of 2026-11-07.

1. **Base branch:** the model's `main`, into which Spencer merged moist V2, the latent-heating version of the moist model, on 2026-11-07, as commit `0dc0627`.

2. **Formulation:** every term takes its physically correct spherical form. Where physics does not fix the form, Section 2 says so and gives the options.

3. **Grid:** uniform in $\sin\varphi$, pole to pole. At the production grid spacing of 39.4 km this is 324 intervals, 0.40 of the production grid's 800, with a latitude spacing of 0.35° at the equator widening to 6.4° at the poles.

4. **Planet:** Earth's mean radius, $a = 6371$ km, and Earth's rotation rate, $\Omega = 7.292115 \times 10^{-5}$ s^{-1}, the value of the WGS84 geodetic system. At that rate one rotation relative to the fixed stars takes 86,164.1 s; the stars define an inertial frame, as the Coriolis force requires. The 86,400 s solar day is 236 s longer, because Earth also advances 0.99° a day along its orbit, in the same sense as it spins, and must turn that much further to face the Sun again. puffins' `ROT_RATE_EARTH` is $2\pi/86{,}400$ s, the solar-day value, 0.27% smaller. Then $2\Omega/a = 2.29 \times 10^{-11}$ m^{-1} s^{-1}, and the SWM's default β of 2×10^{-11} is 13% below it, so every beta-plane run used for comparison is a new run at $\beta = 2\Omega/a$.

5. **The eddy momentum flux divergence $\mathcal{S}$** (Section 2.2): the divergence of an eddy angular-momentum flux, the only one of three spherical forms of SS09's empirical closure that conserves angular momentum. For a uniform-Ro wind the three differ by up to 17% at 30°.

6. **The smoothing of v** (Section 2.3): the vector Laplacian. SS09 added the smoothing "to reduce numerical noise", so no physical process fixes its form.

Open, for Spencer. One choice I would make unless Spencer objects: the forcing on the sphere is the spherical profile that all three of the SWM's forcing profiles were adapted from (Section 2.4), so it needs no decision.

The approach. The spherical model re-discretizes the current scheme and keeps every numerical choice in it: leapfrog stepping with the Asselin filter, the staggered grid with v on cell faces, the monotonized-central (MC) stencil of the eddy term, first-order upwind meridional advection, and both Heaviside gates. Each operator is written so that, with $c = 1$ and the metric term off, it performs the same floating-point operations in the same order as the current one, apart from multiplications by exactly 1.0, which return their operand unchanged. A run in that flat configuration must then reproduce, bit for bit on the same machine, the output of the unmodified model in the four regression configurations, which checks the restructured code before any curvature enters. (The stored regression baselines themselves are reproduced only to cross-machine roundoff, a relative 10^{-11}, so the bit-for-bit reference is the unmodified model run beside the new one.) Any departure from the current numerics is tested separately, one at a time: in the June 2026 rewrite of the numerics, three untested changes bundled into one plan all had to be reverted.

Phases.

1. **Specification.** The operators term by term (Section 2), their discretization including the polar cells (Section 3), and a file-by-file change list (Section 4). *Verified by* a symbolic check of every continuous claim and a numerical check of every discrete one. Complete with this revision.

2. **numpy reference.** Geometry configuration, metric arrays, operators, output, restart files and the command line. *Verified by* unit tests per operator, among them exact discrete conservation of water and exact mirror parity, and by the flat configuration reproducing the unmodified model bit for bit on the same machine.

3. **Validation runs on berimbau**, each with pass criteria set before it runs (Section 5): a drag-limited angular-momentum identity with the eddy term and vertical advection off; steady budgets; the timestep ceiling and a resolution twin; and the scaling of the difference between the sphere and a beta-plane twin with the width of the cell. Every run uses Earth's a and Ω.

4. **numba kernel.** The fused day kernel with the spherical operators. *Verified by* bitwise parity with the numpy reference, through the existing parity suite.

5. **Science runs,** in a separate research project, which takes over once phases 2 to 4 are done (Section 6).

6. **Later, optional.** Moist V1 and V2 and the seasonal forcing on the sphere, verified as in phases 2 to 4.

Cost. The spherical grid has 0.40 of the production grid's points, so a run should cost no more per model day than the current 0.086 s on berimbau. That cost is an estimate pending phase 3's smoke timing: the metric arithmetic adds work per point, and the timestep ceiling near the poles is unmeasured. For development, the closest precedents in this model are the staggered grid and the numba backend, which took about two days of sessions each, 2026-07-11 to 07-12 and 2026-07-13 to 07-14. Phases 2 to 4 together are comparable to both combined, plus the polar cells, which have no precedent here. That comparison is a precedent only; no part of this work has been timed.

2 The spherical equations, term by term

For each term, this section gives the current beta-plane form, as written in SS09 with the correction of Sobel and Schneider (2013, hereafter SS13) and as implemented in the model; the spherical form; its source; and whether physics fixes it. Every spherical form follows from four identities in $y = a \sin\varphi$, each checked symbolically:

$$\zeta = -\frac{\partial(cu)}{\partial y}, \qquad \nabla \cdot v = \frac{\partial(cv)}{\partial y}, \qquad \frac{1}{a}\frac{\partial}{\partial \varphi} = c\,\frac{\partial}{\partial y}, \qquad \frac{u \tan\varphi}{a} = \frac{y\,u}{a^2 c}, \tag{4}$$

for relative vorticity $\zeta = -(a\cos\varphi)^{-1}\partial_\varphi(u\cos\varphi)$ and divergence $\nabla \cdot v = (a\cos\varphi)^{-1}\partial_\varphi(v\cos\varphi)$. Grid cells of equal width in y have equal area, so the area-weighted budgets of mass, heat and water take the same discrete form as on the plane.

2.1 Zonal momentum

Coriolis force and meridional advection. Beta plane: $-v(\beta y - \partial_y u)$. Sphere: $-v(f+\zeta) = -v[f - \partial_y(cu)]$, the meridional flux of absolute vorticity. Source: Xian and Miller's Eqs. (3) and (4), the advection of angular momentum $M = a\cos\varphi\,(\Omega a\cos\varphi + u)$ by v, which is this term written for M. *Fixed by physics.* The model keeps its split into a Coriolis term and first-order upwind advection, the latter now of cu.

Vertical advection of zonal momentum. Beta plane: $-\mathcal{G}\,u\,\partial_y v$, where mass entering the upper layer from below carries zero zonal velocity. Sphere: $-\mathcal{G}\,u\,\partial_y(cv)$. Source: Xian and Miller's Eq. (4), in which rising air carries the lower layer's angular momentum $M_B = \Omega a^2\cos^2\varphi$, that is, zero relative angular momentum; since $M - M_B = a\cos\varphi\,u$, the term for u is the one above. *Fixed by physics.* Both gates apply unchanged: Xian and Miller's gate is SS09's kinematic one, on where the divergence is positive.

Rayleigh drag. $-\epsilon_u u$ on both, a local term. *Fixed.*

2.2 The eddy momentum flux divergence

Beta plane: $\mathcal{S} = v_d\,\mathcal{H}(u)\,\mathrm{sgn}(y)\,\partial_y u$ (SS09 Eq. 2.5), which SS09 describe as "empirical, motivated by the eddy momentum flux divergence in the idealized GCM integrations of Schneider and Bordoni (2008)". *Not fixed by physics.* On the plane the term has two readings, and on the sphere each reading gives a different form, beside a third, literal one:

(a) **A closure on the eddy vorticity flux.** By the Taylor identity, the eddy momentum flux divergence is minus the eddy flux of relative vorticity for nondivergent eddies, on the plane and the sphere alike. SS09's term sets that flux to $v_d\,\mathcal{H}(u)\,\mathrm{sgn}(y)$ times the mean relative vorticity, and the same statement on the sphere gives

$$\mathcal{S} = -v_d\,\mathcal{H}(u)\,\mathrm{sgn}(y)\,\zeta = v_d\,\mathcal{H}(u)\,\mathrm{sgn}(y)\,\frac{\partial(cu)}{\partial y}.$$

Form (a) keeps the drag-free steady closure $\mathrm{Ro} = v/(v+v_d)$ exact.

(b) **The literal transcription,** with ∂_y read as a derivative along meridional distance:

$$\mathcal{S} = v_d\,\mathcal{H}(u)\,\mathrm{sgn}(y)\,c\,\frac{\partial u}{\partial y}.$$

That closure becomes $\mathrm{Ro} = [v - v_d u/(2\Omega a c)]/(v+v_d)$.

(c) **The divergence of an eddy angular-momentum flux.** SS09's term is, away from the equator, the beta-plane divergence of the eddy momentum flux $\overline{u'v'} = v_d\,\mathcal{H}(u)\,\mathrm{sgn}(y)\,u$. The spherical divergence of that flux is

$$\mathcal{S} = \frac{1}{a\cos^2\varphi}\frac{\partial}{\partial\varphi}\left(\cos^2\varphi\;\overline{u'v'}\right) = \frac{v_d\,\mathcal{H}(u)\,\mathrm{sgn}(y)}{c}\,\frac{\partial(c^2u)}{\partial y}.$$

Because $c\,\mathcal{S}$ is then an exact derivative in y, the global torque it exerts reduces to boundary values, which vanish where $u = 0$ bounds the westerlies, just as on the plane. The closure becomes $\mathrm{Ro} = [v + v_d u/(2\Omega a c)]/(v+v_d)$.

For a uniform-Ro wind, forms (b) and (c) equal form (a) times $1 \pm \tan^2\varphi/2$: a difference of 2% at 10°, 7% at 20° and 17% at 30°. *Chosen by Spencer, 2026-11-07:* (c). Eddies redistribute angular momentum and cannot create it, and (c) is the only form that respects that; the exact closure of (a) is a convenience for theory. The reading of SS09's term as a vorticity closure in (a), and as a flux divergence in (c), are mine; SS09 give neither.

2.3 Meridional momentum

Tendency, Coriolis, metric and pressure-gradient terms. Beta plane: $2\partial_t v + \beta y u = -(gH/T_0)\,\partial_y T$. Sphere: $2\partial_t v + (f + u\tan\varphi/a)\,u = -(gH/T_0)\,c\,\partial_y T$. Source: Xian and Miller's Eq. (9), from integrating the vertical derivative of the meridional momentum equation over the troposphere, with the lower layer's meridional wind equal and opposite to the upper layer's and the surface zonal wind negligible. SS13 add that the factor of 2 requires the two layers to be equally thick. The advection $v\,\partial v/\partial y$, which SS13 removed, cancels exactly between the two layers when the lower layer's meridional wind is minus the upper layer's, on either geometry; that reason is my own derivation. The metric term $u^2\tan\varphi/a$ comes from the upper layer alone. *Fixed by physics*, given SS09's two-layer assumptions.

The smoothing of v. Beta plane: $k_v\,\partial_y^2 v$, which SS09 added "to reduce numerical noise". *Not fixed by physics*, since the term represents no physical process. Two candidates: (a) the meridional component of the vector Laplacian, which for any axisymmetric field is the meridional gradient of the divergence, since the rotational part $\mathbf{k}\times\nabla\zeta$ is then purely zonal: $k_v\,c\,\partial_y[\partial_y(cv)] = k_v[\nabla^2 v - v/(a^2\cos^2\varphi)]$; (b) the scalar Laplacian applied to v as though it were a scalar, the literal transcription of $\partial_y^2 v$, which drops the curvature term. *Chosen by Spencer, 2026-11-07:* (a). The vector Laplacian is a vector operator, so its form does not depend on the coordinate, and it acts on the divergence, which is where the grid-scale noise appears. Whether it damps that noise as well as the current form does is a hypothesis, to be tested against the model's v-smoothness monitor and its timestep ceiling.

2.4 Thermodynamics

Adiabatic warming and cooling. Beta plane: $(\delta\Delta_z/H)\,\partial_y v$. Sphere: $(\delta\Delta_z/H)\,\partial_y(cv)$, from the spherical continuity equation, Xian and Miller's Eq. (6). *Fixed by physics.* SS09's thermodynamic equation has no horizontal advection of θ, unlike Xian and Miller's Eq. (12); that is SS09's modeling choice, independent of the geometry, and it is kept.

Newtonian cooling. $(\theta_E - \theta)/\tau$ on both, a local term. *Fixed.*

The forcing θ_E. The SWM's three profiles are beta-plane renderings of one spherical profile, that of Schneider and Bordoni's (2008) idealized GCM, $\hat{T}_{se} - \Delta_h(\sin^2\varphi - 2\sin\varphi_0\sin\varphi)$ for a heating maximum at φ_0. The model's seasonal profile, and its $\sin^2$ profile with no offset, are that profile with $\sin\varphi$ replaced by $\sin(\pi y/2y_1)$, so that y_1 stands in for the pole. The parabolic profile of SS09 is its small-angle form, $\sin^2\varphi \approx (y/a)^2$, with y_1 in the place of a. On the sphere the three therefore collapse to $\theta_E = \theta_{00} - \Delta_y(\sin^2\varphi - 2\sin\varphi_0\sin\varphi)$, with Δ_y the equator-to-pole drop for $\varphi_0 = 0$, as SS09 define it. In the coordinate $y = a\sin\varphi$ that profile is $\theta_{00} - \Delta_y(y^2 - 2y_0 y)/a^2$, with $y_0 = a\sin\varphi_0$: SS09's parabolic profile with $y_1 = a$, exact on the sphere, and with no plateau, since y ends at $\pm a$. The half-width y_1 and the parabolic profile's plateau have no other spherical counterpart. That spherical forcing is the choice I would make unless Spencer objects.

Eddy heat diffusion. Beta plane: $\kappa_\theta\,\partial_y^2\theta$. Sphere: $\nabla\cdot(\kappa_\theta\nabla\theta) = \partial_y(\kappa_\theta\,c^2\,\partial_y\theta)$, which follows from the identities (4). *Fixed*, as the diffusion of a scalar.

2.5 Moisture, for moist V1 and V2

Beta plane: $\partial_t W + \partial_y[-(2a_W - 1)\,vW - D\,\partial_y W] = E_0 - P$, for column water vapor W with lower-layer fraction a_W, eddy diffusivity D, evaporation E_0 and precipitation P. Sphere: $\partial_t W + \partial_y\{c[-(2a_W-1)\,vW - D\,c\,\partial_y W]\} = E_0 - P$, the spherical divergence of the same flux. Precipitation and the latent heating of moist V2 are local. *Fixed by physics.*

2.6 The poles

Xian and Miller (2008) and Adam and Paldor (2009) both run pole to pole, with the meridional wind zero at the poles. On the staggered grid, the outermost u and θ points sit at the poles. There $u = 0$ is the regularity condition, since the relative vorticity contains $u\tan\varphi/a$ and stays finite only if u vanishes at the poles, and it coincides with the wall condition the model already imposes. The meridional wind stays on interior faces, and since $c = 0$ at the poles, no flux crosses them.

2.7 Diagnostics

The local Rossby number becomes $\mathrm{Ro} = -\zeta/f$ with the spherical ζ. Jet and cell-edge latitudes are $\arcsin(y/a)$. The output keeps y in meters as its coordinate, so that existing analysis code reads the files unchanged, and adds latitude beside it.

3 Discretization on the grid uniform in sin(latitude)

Every operator below is transcribed from the model's code at the tip of the moist V2 branch, commit `6503a62` of `feature/moist-v2-sweep`, with the function it modifies named. The merge into `main`, commit `0dc0627`, changed none of these functions: its changes to `ss09/sw_model.py` are confined to the moist V2 relaxation target, in `radiative_target_profile` and the model's constructor.

3.1 The grid and its geometry arrays

The spherical model keeps the staggered layout of the current one: u, θ and the column water vapor W at n_y cell centers, v at the $n_y - 1$ faces between them, and the outermost centers on the boundaries, which are now the poles, $y = \pm a$. At the production equatorial spacing this gives $n_y = 325$ centers and $\Delta y = 2a/324 = 39.33$ km. Since the coordinate is $a\sin\varphi$, Δy is the distance between centers only at the equator. Elsewhere the distance is $\Delta y/\cos\varphi$: 45.4 km at 30° and 78.7 km at 60°. The outermost interior centers sit at 83.63°, and the outermost faces at 85.50°.

The centers are $y_k = a\,(k/m)$ for the integers $k = -m, \ldots, m$, with $m = 162$. The model's configuration builds its grid with `np.linspace`, which is exactly antisymmetric about the equator for the production beta plane ($n_y = 801$) but not on the sphere: `np.linspace(-a, a, 325)` puts 248 of its 325 points off minus their mirror images, by up to 1.9×10^{-9} m. The model's exact mirror parity, under which a hemispherically symmetric forcing keeps u and θ exactly symmetric and v exactly antisymmetric, requires an exactly antisymmetric y, as the comment on `y_v` in `sw_config.py` states. The values $a\,(k/m)$ are antisymmetric by construction, since negating k negates the quotient exactly, and the endpoints are exactly $\pm a$.

The configuration computes these geometry arrays once:

- $c = [(1 - y/a)(1 + y/a)]^{1/2}$ at the centers and the faces, exactly zero at the pole centers and exactly symmetric;
- c^2 at the centers;
- $1/c$ at the interior centers, set to zero at the two pole centers, whose u tendency the condition $u = 0$ there discards;
- $f = (2\Omega/a)\, y$ at the centers and the faces;
- the metric factor $\tan\varphi/a = y/(a^2 c)$ at the faces.

On the beta plane the same arrays are $c = c^2 = 1/c = 1$ exactly and $f = \beta y$, and the metric term is skipped. A multiplication by exactly 1.0 returns its operand unchanged in IEEE arithmetic, so every operator below then reproduces the current one bit for bit. The checks script tests this on prototype operators built from the model's own functions. On 20 random states with exact and signed zeros planted, for both gates on the vertical advection, and with the eddy heat diffusion off and on, every prototype tendency of u, v, θ and W equals the model's bit for bit. Two plausible slips both fail that test: the smoothing computed in an algebraically equal but re-associated form, and the factor of c in the pressure gradient applied after the division by T_0.

3.2 The operators

Notation, for a face field g and a center field q, with the pole centers at $k = \pm m$ and the faces at half-integer indices:

- $\mathrm{div}_k[g] = (g_{k+1/2} - g_{k-1/2})/\Delta y$ at the interior centers, and $2g_{-m+1/2}/\Delta y$ and $-2g_{m-1/2}/\Delta y$ at the two poles (`v_divergence_at_centers`). Each pole value is the finite-volume divergence of a half cell with one face, and since cells of equal width in y have equal area, div is the finite-volume divergence on the sphere.
- $\bar{g}_k$ and $\bar{q}_{k+1/2}$: the two-point averages between the grids (`v_faces_to_centers`, and the face average of u in `dv_dt`), with $\bar{v} = 0$ at the poles.
- $\delta_k^{\mathrm{up}}[q]$: the first-order upwind difference of q on the sign of $\bar{v}_k$ (`merid_advec_u`).
- $M_k[q]$: the MUSCL derivative with monotonized-central limited slopes, upwinded from the equatorward side (`muscl_mc_du_dy`).
- $L_{k+1/2}[g]$: the three-point face Laplacian in the mirror-symmetric association, with mirror ghosts beyond the outermost faces (`v_face_laplacian`).

Coriolis force and meridional advection of u (`du_dt`, `merid_advec_u`). Now $\beta y_k\, \bar{v}_k - \bar{v}_k\, \delta_k^{\mathrm{up}}[u]$; on the sphere $f_k\, \bar{v}_k - \bar{v}_k\, \delta_k^{\mathrm{up}}[cu]$.

Vertical advection of u (`vert_advec_u`). $\mathcal{G}_k\, u_k\, \mathrm{div}_k[cv]$, with the kinematic gate reading $\mathrm{div}_k[cv]$.

The eddy term, form (c) (`edd_mom_flux_div_u`). $\mathcal{S}_k = v_d\,\mathcal{H}(u_k)\,\mathrm{sgn}(y_k)\,c_k^{-1} M_k[c^2 u]$. The MUSCL derivative is a difference of reconstructed face values: in the north, $M_k[q] = (q^L_{k+1/2} - q^L_{k-1/2})/\Delta y$, with $q^L_{k+1/2} = q_k + \frac{1}{2}\Delta y\,\sigma_k$ the reconstruction from the equatorward cell and σ the limited slope, and in the south the mirror form. So $c_k \mathcal{S}_k\,\Delta y$ is a difference of face values within each hemisphere's westerly region, and the sum over that region equals v_d times its boundary values, as in the continuous form. The checks script confirms this for $u = U\sin\varphi\cos\varphi$ to a relative 5×10^{-17}, and the literal form (b) fails the same check. The limiter acts on c^2u, the quantity form (c) differentiates, so the existing function is reused unchanged. The alternative, limiting u and multiplying its face values by c^2, also reduces to the current operator on the plane, and it is the first change to test if the spherical eddy term proves noisy.

Meridional momentum (`dv_dt`). At each face,

$$2\,\partial_t v = -\Big(f + \frac{\tan\varphi}{a}\,\bar{u}\Big)\bar{u} - \frac{gH}{T_0}\,c\,\frac{T_{k+1} - T_k}{\Delta y} + k_v\,c\,L[cv].$$

The smoothing $c\,L[cv]$ is algebraically $c\,(\mathrm{div}_{k+1}[cv] - \mathrm{div}_k[cv])/\Delta y$, the face gradient of the discrete divergence, because the mirror ghost reproduces the half-cell divergence at the pole. The smoothing is therefore the discrete form of the chosen vector Laplacian, and it is exactly dissipative: summed over the faces, v times $c\,L[cv]$ equals minus the sum over centers of $\mathrm{div}[cv]^2$, with half weights at the two pole half cells. So the smoothing never increases the sum of v^2 over the faces, and lowers it whenever the divergence is nonzero. The checks script confirms both identities to relative differences of 3×10^{-16} and 1×10^{-16}. With $c = 1$ they hold for the current smoothing too, which is already the gradient of the discrete divergence.

Thermodynamics. The adiabatic term, in `vert_advec_theta`, is $-(\delta\Delta_z/H)\,\mathrm{div}_k[cv]$. The eddy heat diffusion, in `eddy_heat_flux` and off by default, becomes $\kappa_\theta\,\partial_y(c^2\partial_y\theta)$, with the current centered `np.gradient` applied twice.

Moisture (`moisture_transport_tendency`). The face flux is $F = c\,[\,c_W W^{\mathrm{MC}} - D\,c\,(W_{k+1} - W_k)/\Delta y\,]$, with $c_W = -(2a_W - 1)v$ the column transport velocity and W^{MC} the current limited upwind face value; the tendency is $-\mathrm{div}_k[F]$. Total water is conserved exactly, since the flux vanishes at the poles, and the half-cell weights of `cwv_integral` are the equal-area weights. The checks script confirms this to a relative 3×10^{-18}.

Local terms. Rayleigh drag, Newtonian cooling, precipitation and the latent heating of moist V2 are unchanged.

Forcing (`theta_e.py`). A new profile, $\theta_{00} - \Delta_y(s^2 - 2s_0 s)$ with $s = y/a$ and $s_0 = \sin\varphi_0$. Moist V2 relaxes to this profile itself, as it relaxes to the current profiles since the merge; an explicit radiative target, from `--delta-y-rad`, replaces Δ_y with Δ_y^{rad}, as now.

3.3 The polar cells

Each pole center sits at the edge of a half cell covering 0.154% of the sphere's area, 1/648. There $u = 0$, as at the current walls. No flux crosses a pole, since $c = 0$ there, so the pole divergence is that of a half cell with one face. The eddy term is zero at the poles, through $1/c = 0$; the condition $u = 0$ discards the pole tendency anyway, and a zero keeps u's leapfrog history there finite.

Near a pole, a regular axisymmetric field is even or odd in the colatitude χ. Since $y/a = \cos\chi$ and χ^2 is an analytic function of $1 - y/a$, an even field is a smooth function of y up to the pole, while an odd field is $(a-y)^{1/2}$ times a smooth function. The winds u and v are odd, and so is c^2u; cu, cv, θ, T and W are even. The divergence, the pressure gradient, the advection of cu, the smoothing and the moisture fluxes all act on even fields, so the uniform grid in y resolves them up to the poles. Three operations act on odd fields and lose accuracy in the outermost cells: the averages of u and v between centers and faces, and the eddy term's derivative of c^2u. For the regular field $v = V\sin\varphi\cos\varphi$, the face-to-center average of v is off by 3.5% at 83.6°, 0.85% at 81.0° and 0.22% at 77.2°, while within 60° of the equator its error stays below 10^{-4} of $\max|v|$. The unit tests of phase 2 measure each operator's convergence order on regular test fields over the whole domain, polar cells included.

3.4 The timestep

In the coordinate y, gravity waves travel at c times their physical speed, because the pressure gradient and the divergence each carry one factor of c. With the spacing in y uniform, the wave limit on the timestep is then set at the equator and equals the beta plane's at the same Δy. The advection speed in y is cv, and the smoothing's rate carries c^2, so neither grows toward the poles. The one term that does grow is the metric frequency $u\tan\varphi/a$: at the outermost faces, where $\tan\varphi = 12.7$, it is 2.0×10^{-5} s^{-1} for $u = 10$ m s^{-1} and 6.0×10^{-5} s^{-1} for 30 m s^{-1}, 0.14 and 0.41 of 2Ω. These are expectations from the equations: phase 3 finds the ceiling by bisection, and no run configuration is chosen from them before then.

4 Changes, file by file

First written against commit `6503a62`, and re-derived on 2026-11-08 against the merged `main`, commit `0dc0627`, with no change to the list. Between the two commits, `ss09/sw_config.py` changed two moist defaults, the lower-layer water fraction a from 0.85 to 0.79 and the critical column water vapor W_c from 50 to 40 kg m^{-2}, and made the radiative contrast Δ_y^{rad} optional where it had defaulted to 75 K; `ss09/sw_model.py` changed the moist V2 relaxation target; `ss09/cli.py` changed help text; and the tests gained a fourth regression baseline, for moist V2, with its numba counterpart. None of these changes touches a field, function or option the list modifies, and the other source files in the list are unchanged.

- `ss09/sw_config.py`: fields `geometry` (`beta_plane`, the default, or `sphere`), `radius` (6.371×10^6 m) and `rot_rate` (7.292115×10^{-5} s^{-1}); on the sphere, the grid a (k/m) with n_y odd, the domain fixed at $2a$ (an explicit `domain_size` or `beta` refused), the staggered grid only; the geometry arrays, with their flat values on the beta plane.

- `ss09/sw_model.py`: the operators above, through the geometry arrays; the collocated layout refuses the sphere, since it is kept frozen to reproduce an earlier published version of the model; the output's y described as $a\sin\varphi$ on the sphere, with latitude coordinates for the centers and faces.

- `ss09/numba_backend.py`: `run_day` takes the geometry arrays and a switch for the metric term, and each fused loop gains the same factors in the same positions as the numpy reference. The shared difference array `diff` splits in two on the sphere, one for cu and one for c^2u, which on the plane both equal the current one exactly.

- `ss09/theta_e.py`: the spherical profile, with φ_0 in degrees. The existing profiles are refused on the sphere, since their y_1 and plateau have no meaning there. Seasonal migration of φ_0 waits for the later phase.

- `ss09/hadley_diagnostics.py`: the local Rossby number computed from $\partial_y(cu)/f$ in place of $\partial_y u/(\beta y)$, with the call site passing f and c in place of β; jet, cell-center and cell-edge positions also reported as latitude. A maximum of u in y is a maximum in latitude, since y increases with φ, so the positions' definitions are unchanged.

- `ss09/daily_results.py`: the latitude coordinates and the Rossby number's description.

- `ss09/restart_state.py`: the compatibility check compares `geometry`, `radius` and `rot_rate`, refusing a beta-plane restart for a spherical run.

- `ss09/cli.py`: `--geometry`, `--radius`, `--rot-rate`, the spherical forcing type and `--lat-0`; on the sphere, `--beta`, `--domain-size`, `--y0` and `--y1` refused.

- `ss09/steady_state.py`: unchanged. Its kinetic-energy metric, the mean of $u^2 + v^2$ over the centers, is already an equal-area mean apart from counting the two pole half cells as whole cells, and the smoothness monitor looks for grid-scale noise in the grid's own coordinate.

- Tests: new tests of the geometry (antisymmetry, arrays, flat reduction) and of the operators (convergence order on regular test fields, the dissipation identity, water conservation, the eddy torque, mirror parity over multi-day runs); the numba parity suite extended to spherical configurations; the regression tests unchanged, which since the merge cover four configurations: collocated, staggered, moist V1 and moist V2.

- Documentation: a section of `SCIENCE.md` on the spherical formulation, and the new options in `README.md` and `CLAUDE.md`.

5 Validation, with pass criteria

Every criterion is set before its run. Where a criterion tests the model's operators, an analysis script in this repository's `validation/` directory computes it, independently of those operators.

Code checks, on one machine (phases 2 and 4).

1. The flat configuration reproduces the unmodified model bit for bit in the four regression configurations, against a reference run, on the same machine, of the `main` commit the spherical branch starts from. Commit `6503a62` no longer serves, since the merge changed the moist defaults and added the moist V2 configuration.

2. The unit tests above.

3. The numba kernel reproduces the numpy reference bit for bit on the sphere, across the existing matrix of physics switches.

4. A hemispherically symmetric forcing keeps exact mirror symmetry over a 30-day spherical run, and a run continued from a restart file reproduces the uninterrupted run bit for bit.

Runs on berimbau (phase 3), all with the spherical forcing at $\Delta_y = 50$ K and $\varphi_0 = 0$ unless stated, each plotted and inspected before any number from it is reported.

(i) **Drag-limited angular-momentum identity.** With $v_d = 0$ and the vertical advection off, the steady u equation is $\bar{v}\,(f+\zeta) = \epsilon_u u$, so $1 - \mathrm{Ro}_{\mathrm{loc}} = \epsilon_u u/(f\bar{v})$ at every point, for $\mathrm{Ro}_{\mathrm{loc}} = -\zeta/f$. Runs at $\epsilon_u = 10^{-8}$ and 10^{-9} s^{-1}, the second at five drag timescales, 58,000 model days. *Pass:* between 5° and the jet, the two sides agree to within the residual of the steady u budget, with ζ computed by the analysis script from u. A model missing the metric part of the advection would leave a residual of $u \tan\varphi/(af)$: 0.057 for $u = 50$ m s^{-1} at 20°, and 0.022 for 20 m s^{-1} at 10°.

(ii) **Steady budgets** at the production parameters. *Pass:* the residuals of the u, v and θ budgets from the final 30-day means, relative to each budget's largest term, are no larger than the beta-plane twin's of item (iv).

(iii) **Timestep and resolution.** The timestep ceiling at $n_y = 325$, found by bisection, and a twin at $n_y = 649$, with Δy halved. *Pass:* the whole-field $\max|\Delta|$ of u, v and θ within 40° of the equator, each scaled by its field's maximum, is no larger than that of the beta-plane twin at the same two spacings. The polar cells are reported separately.

(iv) **Scaling of the curvature effect.** The sphere beside its beta-plane twin: $\beta = 2\Omega/a$, walls at $\pm\pi a/2$, the distance from equator to pole, and the same θ_E as a function of latitude, with $\varphi = y/a$. That forcing is the existing profile of Schneider and Bordoni (2008) with $y_1 = \pi a/2$ and $y_0 = a\varphi_0$. The two models then differ only through curvature: the factors of c, the metric term, $2\Omega\sin\varphi$ in place of $2\Omega\varphi$, and the spherical form of the eddy closure. Each enters at relative order φ^2, and the square of the cell's latitude extent grows with Δ_y, so the relative difference between sphere and twin in the jet and cell-edge latitudes should scale with the square of the twin's cell-edge latitude, φ_e^2 in radians. Runs at $\Delta_y = 12.5$, 25 and 50 K, with the twin on 508 intervals ($n_y = 509$, keeping a center on the equator), a spacing 0.18% above the sphere's. *Pass:* that relative difference divided by φ_e^2 is the same across the three rungs to within the numerical uncertainty from item (iii), once the next-order term in φ_e^2 is allowed for. A curvature term entering at the wrong order, such as a dropped factor of c, breaks the proportionality. The runs belong to phase 3; the next-order term comes from theory, so this criterion is evaluated in the science project of Section 6. This test replaces the earlier plan's ladder of planetary radii at fixed $2\Omega/a$: at fixed Δ_y that ladder narrows the cell in latitude in the same way, and it would have required planets other than Earth.

6 Handoff to the science runs

Phases 2 to 4 belong to this repository. Phase 5, the science runs on the sphere, belongs to a separate research project in the private repository `spencerahill/fixed-ro-eq-area`. That project takes over once both of the following hold:

1. The spherical option is merged into `main`, with code checks 1 to 4 of Section 5 passing.

2. Validation runs (i) to (iii) have passed, and the runs of item (iv) are complete, with the evaluation of their pass criterion left to the science project.

The science project then needs four things from this repository: the commit of `main` that adds the spherical option; the location on berimbau of the output of validation runs (i) to (iv); the analysis scripts in `validation/` that scored items (i) to (iii); and the measured timestep ceiling at $n_y = 325$ with the cost per model day on the sphere.

7 Risks

1. **The polar cells.** The averages between grids and the eddy term lose accuracy there, as the polar-cells subsection quantifies. If grid-scale noise appears near the poles, in the v-smoothness monitor or the resolution twin, the first remedy to test is a stronger smoothing confined to the outermost cells, a departure from the current numerics tested on its own.

2. **The limiter on c^2u.** If the gated eddy term is noisy on the sphere, the alternative of limiting u is the first change to test.

3. **The timestep.** The expectation that the ceiling equals the beta plane's at the same Δy comes from the equations, and the bisection may contradict it.

4. **The bit-for-bit reduction.** The numba kernel shares one difference array between two terms that need different ones on the sphere. Splitting it keeps the flat values exactly, but any reordering made while splitting would show as a failure of code check 1, and the fix belongs in the restructured code.

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